**LUC@BJTU Weihai Campus**

**Fundamentals of Communication Systems (CNSCC160) Tutorial 2**

B **1. What is a signal in the context of communication systems?**

a) A function of multiple dependent variables
b) A function of one or more independent variables containing information
c) A process with no temporal variation
d) A static electrical circuit

A **2. Which form represents a signal as a voltage source?**

a) Thevenin's form
b) Norton's form
c) Kirchhoff's form
d) Coulomb's form

C **3. What type of signal is defined for every instant of time within a range?**

a) Discrete-time signal
b) Pulse signal
c) Continuous-time signal
d) Sine wave signal

x(t) 连续信号

B **4. A discrete-time signal is represented as:**

a) x(t) for continuous values of t
b) x[n], where n is a sequence of numbers
c) y(t) for finite values
d) z[n], for complex numbers only

x[n] 离散信号

D **5. Which of the following is NOT an example of an analogue signal?**

a) Voice
b) Sine wave
c) Video
d) Serial binary code

周期性信号

C **6. A signal pattern that repeats over time is known as a:**

a) Aperiodic signal
b) Discrete signal
c) Periodic signal
d) Modulated signal

B **7. What is the period (T) of a signal related to?**

a) The speed of propagation 传播速度
b) The signal frequency
c) The amplitude of the signal
d) The wavelength of the medium

C **8. The range of frequencies a signal contains is known as the:**

a) Wavelength
b) Period
c) Spectrum 频域
d) Phase

C **9. Which formula is used to obtain a signal's frequency spectrum?**

a) Nyquist formula
b) Shannon's capacity
c) Fourier series
d) Euler's theorem

C **10. What does Signal-to-Noise Ratio (SNR) quantify?**

a) The bandwidth of a channel
b) The amount of signal distortion
c) The ratio of signal power to noise power
d) The total power of a system

D **11. Which of the following is true about a noiseless channel?**

a) It allows zero transmission errors
~~b)~~ It is described by Shannon's capacity

~~c)~~ It cannot transmit high bit rates
d) Its bit rate is determined by the Nyquist formula

$$\text{Bit Rate} = 2 \cdot B \cdot \log_2 L$$

B **12. Which of the following transmission media requires line-of-sight?**

a) Optical fiber
b) Microwave
c) Radio waves
d) Copper wire

C **13. Which type of wave cannot penetrate walls?**

a) Radio waves
b) Microwaves 但是高频 Microwave 也不能穿墙
c) Infrared waves 红外线波
d) Visible light

B **14. What is the purpose of multiplexing in communication systems?**

a) To compress data for easier storage
b) To increase the number of signals transmitted over a single channel
c) To enhance signal quality by amplifying the signal
d) To separate signals by increasing their frequency

D **15. What type of multiplexing uses time slots to transmit multiple signals?**

a) Frequency-division multiplexing (FDM)
b) Code-division multiplexing (CDM)
c) Space-division multiplexing (SDM)
d) Time-division multiplexing (TDM)